SOLAR DATA LOGGER
Formula & Analytics Profile

Technical specification | Dashboard metrics, calculations, alerts, customer profile and reporting

Reference example (from supplied screenshots): 5 kWp DC plant; 412.5 kWh monthly generation; 22 elapsed/assumed production days; 4.5 peak-sun-hours/day; tariff ₹8/kWh; project cost ₹300,000; emission factor 0.82 kg CO_2/kWh. These are example inputs, not verified live measurements.

1 | Key corrections to the supplied formulas

The screenshot computes expected generation as 5 × 4.5 × 22 = 495 kWh, not 493 kWh. With 495 kWh, the displayed ratio becomes 83.33%; using 493 kWh yields 83.67%. The label “performance ratio” is only an approximate generation-vs-benchmark index when using this simple peak-sun-hours model. A standards-based solar PR uses plane-of-array irradiation and reference yield.

412.5 × ₹8 = ₹3,300; 412.5 × 0.82 = 338.25 kg CO_2; 412.5 / 5 = 82.5 kWh/kWp. For a 31-day month, CUF = 412.5 / (5 × 744) × 100 = 11.09%. ₹39,600 annual savings is an annualization of ₹3,300 × 12, not a measured annual total. Simple payback = ₹300,000 / ₹39,600 = 7.58 years.

Important financial distinction: generation × retail tariff overstates monetary savings if some solar energy is exported at a lower credit rate. Calculate self-consumption and export separately whenever smart-meter data is available.

2 | Input fields and data sources

Field	Unit / source	Notes
Timestamp; timezone	ISO date/time; logger	Track freshness and local day boundaries
PV DC nameplate capacity	kWp; installation master	DC STC capacity, not inverter AC kW
Inverter AC power / energy	kW / cumulative kWh; inverter	Energy from positive cumulative deltas
PV irradiance / PSH	kWh/m²; sensor or weather API	Plane-of-array preferred for true PR
Grid import / export	kWh; bidirectional meter	Separate cumulative registers
Load consumption	kWh; meter or derived	Ensure synchronized intervals
Tariff and export credit	₹/kWh; customer contract	Time-of-use rates if applicable
Temperature, alarms, connectivity	°C, codes, heartbeat; inverter	Vendor-specific thresholds
Installation date, capex	date, ₹; customer profile	Include subsidies/maintenance as needed

3 | Generation and performance formulas

F01 — Interval generated energy

Formula: E_interval (kWh) = max(0, E_total_end − E_total_start)

Use inverter lifetime energy counter if reliable; handle resets, rollovers, missing samples and duplicates. Alternative: sum(P_ac_kW × Δt_hours) for valid intervals.

F02 — Today / yesterday / month / year

Formula: E_period = Σ valid interval generation within local period

Use actual calendar boundaries and local timezone; label incomplete periods as “so far”.

F03 — Expected generation (simple benchmark)

Formula: E_expected = P_DC(kWp) × PSH(h/day) × days × benchmark_factor

If factor is 1.0, this is a theoretical irradiation-based benchmark, not an engineering prediction. Include temperature, shading and other loss factors for realistic prediction.

Example: 5 × 4.5 × 22 = 495 kWh (factor 1.0).

F04 — Generation achievement (%)

Formula: Achievement = E_actual / E_expected × 100

Show “N/A” if expected energy is zero or missing.

Example: 412.5 / 495 × 100 = 83.33%.

F05 — IEC-style performance ratio (PR)

Formula: PR = Y_f / Y_r × 100; Y_f = E_AC / P_DC; Y_r = H_POA / G_STC

H_POA is plane-of-array irradiation in kWh/m²; G_STC = 1 kW/m². Use aligned time windows and calibrated measurements. Not inverter efficiency.

F06 — Specific yield

Formula: Specific yield = E_AC(kWh) / P_DC(kWp)

Compare sites on the same reporting period.

Example: 412.5 / 5 = 82.5 kWh/kWp.

F07 — Capacity utilization factor (CUF)

Formula: CUF(%) = E_AC(kWh) / [P_DC(kW) × period_hours(h)] × 100

Specify whether DC or AC capacity is used; here DC nameplate. For monthly CUF use all calendar hours in the reporting period.

Example: 412.5 / (5 × 744) × 100 = 11.09%.

F08 — Peak day and average daily generation

Formula: Peak_day = max(E_daily); Average_day = Σ E_daily / count(valid_days)

Avoid treating partially measured days as full days. Screenshot values: today 18.4 kWh; yesterday 17.9 kWh; peak 21.2 kWh.

F09 — Month-on-month / year-on-year change

Formula: Change(%) = (current_period − comparison_period) / comparison_period × 100

Compare same number of elapsed days for month-to-date, or complete matching months; return N/A if baseline is zero.

F10 — Weather-normalized generation

Formula: Normalized yield = E_AC / H_POA; or compare actual to modelled E_expected

Use irradiance-adjusted comparisons to avoid confusing cloudy weather with plant faults.

4 | Grid, consumption and financial formulas

F11 — Self-consumed solar energy

Formula: Solar_self_used = Solar_generation − Solar_export

Valid for a conventional grid-tied plant without battery or other flows; use metered flows and adjust for battery charging/discharging if present.

F12 — Total load consumption

Formula: Load = Solar_self_used + Grid_import

For no-storage system and consistent metering boundaries. Equivalent: Load = Solar_generation + Grid_import − Grid_export.

F13 — Net grid energy

Formula: Net_grid = Grid_import − Grid_export

Positive is net import; negative is net export. Billing may use separate registers and slabs.

F14 — Self-consumption ratio

Formula: Self_consumption(%) = Solar_self_used / Solar_generation × 100

Measures how much solar energy is used on site; not the same as solar self-sufficiency.

F15 — Solar self-sufficiency

Formula: Self_sufficiency(%) = Solar_self_used / Total_load × 100

How much of customer consumption was covered by solar.

F16 — Financial benefit (accurate model)

Formula: Savings = Σ(Self_used_kWh × avoided_import_tariff) + Σ(Export_kWh × export_credit) − additional_charges

Calculate interval-by-interval for TOU billing; account for fixed charges, demand charges, taxes, net-metering rules and settlement credits.

Example: Only if all 412.5 kWh offsets imports at ₹8/kWh: ₹3,300.

F17 — Simple gross generation value

Formula: Gross_value = Generated_kWh × tariff

Useful as an indicative display only; do not automatically call it bill savings.

Example: 412.5 × ₹8 = ₹3,300 indicative.

F18 — Annual savings forecast

Formula: Forecast_annual = Σ 12 modelled monthly savings; rough run-rate = current_month_savings × 12

The rough run-rate assumes identical months and is unsuitable for seasonally variable generation.

Example: ₹3,300 × 12 = ₹39,600 rough run-rate.

F19 — Simple payback

Formula: Simple_payback_years = Net_initial_investment / annual_net_cash_savings

Annual net cash savings should deduct recurring O&M and other costs. Does not account for discounting, degradation or tariff escalation.

Example: ₹300,000 / ₹39,600 = 7.58 years (before O&M).

F20 — Simple ROI / lifetime return

Formula: Annual_simple_ROI(%) = annual_net_benefit / net_initial_investment × 100; Lifetime_net = Σ annual_net_cashflow − investment

For investment analysis add degradation, inverter replacement, inflation, discount rate, NPV and IRR.

Example: ₹39,600 / ₹300,000 × 100 = 13.2% illustrative gross annual ratio.

5 | Environment and reporting

F21 — Avoided CO_2 (estimate)

Formula: CO2_avoided_kg = eligible_solar_kWh × grid_emission_factor(kg CO_2/kWh)

Use region- and year-specific grid factor; distinguish generation-based avoided emissions from exported/consumed accounting; avoid claiming direct measurement.

Example: 412.5 × 0.82 = 338.25 kg CO_2.

F22 — Tree equivalent (communication-only)

Formula: Tree_equivalent = CO2_avoided_kg / assumed_kg_CO2_per_tree_for_defined_period

Tree sequestration varies substantially with species, age and timeframe. State factor and period prominently; screenshot “15 trees” implies 22.55 kg/tree and is not a universal conversion.

F23 — Monthly PDF headline

Formula: Headline = verified bill_savings if meter/tariff data available, else estimated_generation_value

Use “estimated value” rather than “saved” if only inverter generation is available. Include month/year, last updated time and data coverage.

6 | System status, connectivity and intelligent alerts

F24 — Online / offline

Formula: Online = (now − last_valid_heartbeat) ≤ configured_timeout AND communication_status = OK

Use configurable 5–15 minute timeout depending on logger sampling. During planned nighttime sleep, distinguish expected sleep from offline.

F25 — Inverter temperature

Formula: Temp_status = vendor_alarm_status OR temperature > vendor_limit

42°C cannot universally be called OK; compare with inverter model, ambient temperature and manufacturer limits.

F26 — Panel soiling / cleaning alert

Formula: Soiling_loss(%) ≈ max(0, (Expected_weather_adjusted − Actual) / Expected_weather_adjusted × 100)

Flag only after persistent daytime underperformance; exclude clouds, shading, clipping, faults and sensor errors. Prefer clean-reference/string comparison or dedicated soiling sensor. The “2% loss” shown is illustrative, not proven.

F27 — Wind warning

Formula: Wind_alert = forecast_gust_m_s ≥ site_configured_threshold

Use local weather API, gust forecast and engineering limits; notify user, do not automatically advise unsafe rooftop access.

F28 — Faults / wiring / grid status

Formula: Status = vendor inverter fault codes + grid voltage/frequency alarms + device telemetry

Wiring health cannot be confirmed by software status alone; label “no detected wiring alarm”, not “wiring verified safe”.

F29 — Data coverage and freshness

Formula: Coverage(%) = valid_expected_samples / expected_samples × 100; Freshness = now − newest_sample_time

Suppress misleading KPIs or mark estimated when coverage is insufficient. Track API failures, late arrivals, counter resets and timestamp drift.

F30 — Weather-based low-output detection

Formula: Alert if Actual_day < Weather_adjusted_expected_day × (1 − tolerance) for N comparable days

Set tolerance using plant history and quality of weather input; do not alert simply because a cloudy day produced little energy.

7 | Dashboard modules matching the handwritten requirements

1. Update time: last successful logger sync and data timestamp.

2. Today’s energy: daily cumulative delta and last interval power.

3. Performance ratio: distinguish IEC-style PR from benchmark achievement.

4. Period savings: actual self-use + export credits; show estimated if unmetered.

5. Smart meter: total generated, net grid export/import, self-consumed and load.

6. Daily generation bar chart: date vs kWh; missing days flagged.

7. Expected kWh: site capacity, irradiation, weather and system-loss assumptions.

8. Monthly CUF: actual kWh divided by capacity × calendar hours.

9. Online/offline: logger heartbeat, inverter status and last-seen timestamp.

10. Customer details: name, address, inverter make/model, capacity, installation date, customer code, DISCOM, customer number; access-controlled.

11. Graphics: CO_2 estimate, tree-equivalent with stated assumption.

12. Soiling: inferred performance-loss alert, never based solely on low output.

13. Wind: forecast wind/gust on dashboard, severe-wind alert.

14. Last-year same-month comparison: matched monthly kWh and percentage change.

8 | Example dashboard outputs and data validation

Metric	Reference result	Display qualifier
Monthly generation	412.5 kWh	Assumed example month
Expected generation	495 kWh	5 × 4.5 × 22
Achievement vs expected	83.33%	Not IEC PR
Gross energy value	₹3,300	Not necessarily bill savings
CO_2 estimate	338.25 kg	Factor = 0.82 kg/kWh
Specific yield	82.5 kWh/kWp	5 kWp DC
CUF (31-day month)	11.09%	744 hours
Annual gross run-rate	₹39,600	12 × one month
Simple payback	7.58 years	Before O&M; gross basis

9 | Implementation and quality rules

Use explicit units for every field; store raw meter counters, timestamps and source IDs; deduplicate and correct counter resets; keep local timezone and reporting cut-off consistent; validate nonnegative energy; handle zero denominators; version tariff, emission factors and weather assumptions; mark estimated values; protect customer identity; show data provenance and last updated timestamp.

Recommended API objects: plant_profile, inverter_telemetry, logger_heartbeat, interval_generation, smart_meter_import_export, weather_irradiance, tariff_schedule, alerts, daily_aggregates, monthly_aggregates and report_snapshot. Keep raw and derived data separate.

Validation tests: 5×4.5×22 must equal 495; 412.5/495×100 ≈ 83.33%; 412.5×8 = 3300; 412.5×0.82 = 338.25; 412.5/(5×744)×100 ≈ 11.09%; 300000/39600 ≈ 7.58. Test zero production, absent smart meter, partial month, negative counter delta, leap year, DST/timezone boundaries and communication loss.

10 | Reference assumptions and limitations

This profile analyzes user-supplied mockup screenshots and handwritten requirements. No original logger CSV, smart-meter readings, irradiance sensor feed, tariff agreement or inverter manufacturer specifications were supplied. Therefore example status values (HEALTHY, 42°C OK, CLEAN PANELS, 2% soiling, etc.) cannot be independently verified. Formula conventions align broadly with established PV performance terminology; site-specific engineering validation is required before production deployment.